Problem 1

Write a function `introduce(name, age)` that prints:

```
My name is <name> and I am <age> years old.
```

Call it using positional arguments.

Sample Input:

Rahim 28

Sample Output:

My name is Rahim and I am 28 years old.

Problem 2

Write a Python generator function called batch_generator. It should take a list of numbers (representing data samples) and a batch_size. The generator should yield one batch at a time as a list.

Example Input:

data = [1, 2, 3, 4, 5, 6, 7, 8]

batch_size = 3

Expected Output (when iterated):

[1, 2, 3]

[4, 5, 6]

[7, 8]

Problem 3

Given a list `[1,2,3,4]`, use `map()` and `lambda` to create a new list with squares of each number.

Sample Input:
No input

Sample Output:

[1, 4, 9, 16]

Problem 4

Write a function `power(base, exponent)` that returns the result of base raised to exponent. If the exponent is not given, it should calculate the square.

Sample Input:

5

Sample Output:

25

Sample Input 2:

5 3

Sample Output 2:

125

Problem 5 (Conceptual Session)

Given a list `[1,2,3,4,5,6]`, use `filter()` to select even numbers and `map()` with `lambda` to square them.

Sample Input:
No input

Sample Output:

[4, 16, 36]